1.4 Multiplying Multi-Digit Numbers With Decimals to the Thousandths

Lesson Introduction

Benchmark	MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.	Learning Target	• I can fluently multiply multi-digit numbers with decimals to the thousandths.

Benchmark Coherence	Building On	Working Toward
	MA.5.NSO.2.5 Multiply and divide a multi-digit number with decimals to the tenths by one-tenth and one-hundredth with procedural reliability.	MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

Essential Questions	• How do you multiply decimals? • How is it helpful to think about place value and rewriting decimals as fractions when multiplying decimals? • What are the similarities and differences between multiplying decimals with the area model, partial products, and the standard algorithm?
Lesson Narrative	The lesson begins with an open-ended Warm-Up that reviews operations with decimals, followed by a Refresher that spirals learning from 5th grade on reasoning with decimal operations. The lesson then continues by making connections to fractional thinking with base powers of 10, area models, partial products, and the standard algorithm for multiplying decimals to build conceptual understanding and procedural fluency.

Component Summary				
	1.4.1 Warm-Up (~5 min.)	Use place value reasoning to write factors with tenths *(SE p. 20)*	1.4.4 Guided Practice (5-10 min.)	Students use preferred method(s) to multiply decimals to the thousandths *(SE p. 23)*
	1.4.2 Refresher (5-10 min.)	Rewriting decimals as base-10 fractions *(SE p. 21)*	1.4.5 Try It (10 min.)	Students use preferred method(s) to multiply decimals to the thousandths *(SE p. 24)*
	1.4.3 Guided Instruction (10-15 min.)	Using area model, partial products, and standard algorithms *(SE p. 22)*	1.4.6 Your Turn! (5-10 min)	Students use any method(s) to multiply decimals to the thousandths *(SE p. 25)*
	1.4.7 Wrap-Up (~5 min.)	Fluency check for multiplying decimals *(SE p. 25)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms</u>: N/A <u>Additional Terms</u>: • Area model • Partial product • Standard algorithm	• (Optional) Place value chart • (Optional) Chart paper • (Optional) Markers	• (Optional) Prepare chart paper and markers to create anchor charts showing all three methods introduced and practiced in 1.4.3 Guided Instruction and 1.4.4 Guided Practice. Consider posting in a shared classroom space as a reference on procedural steps to multiply decimals to refer back to after the lesson.

<table>
<tr><td rowspan="4">Teacher Tips</td><td colspan="2">Common Misconceptions</td><td colspan="2">Things to Consider</td></tr>
<tr><td colspan="2">Students may struggle with decimal place value, often confusing hundredths and thousandths.
Students may struggle with making sense of the standard algorithm.
Students may struggle with placement of the decimal in the product. Consider building understanding of base-10 thinking, fractional thinking, and estimation for scaffolding strategies as modeled in the 1.4.2 Refresher.</td><td colspan="2">This lesson’s benchmark clarification calls for multiplying multi-digit numbers up to the thousandths with no more than 5 total digits, which are included in the lesson. However, not every problem goes to the thousandths as students are working the problems by hand.
1.4.2 Refresher is designed to make the connection between place value and powers of 10, but additional scaffolding may be needed for some students.
The standard algorithm is the most abstract method of solving. If you notice students struggling, connect it back to a more concrete method, such as area model, to build understanding.
If time is an issue, consider assigning all or part of 1.4.5 Your Turn! for homework.</td></tr>
<tr><td colspan="2">Literacy and Language Routines</td><td colspan="2">Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td colspan="2">Notice and Wonder
1.4.3 Guided Instruction and 1.4.4 Guided Practice give students opportunities to notice similarities and differences between the approaches for multiplying decimals presented in this lesson. Consider applying “Notice and Wonder” routines on each question set, positioning students to discover the relationship between partial products and standard algorithms.</td><td colspan="2">MA.K12.MTR.3.1: Mathematical Fluency
Throughout the lesson, students will be working to fluently multiply multi-digit decimals to the thousandths place using a variety of methods. Consider using 1.4.6 Your Turn! and 1.4.7 Wrap-Up components to provide plenty of practice using the standard algorithm, area model, and partial products to complete multiplication accurately and with confidence.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="4">1.4.6 Your Turn! - Consider specifying which problems students complete by complexity based on student confidence and comfortability with the standard algorithm. For students who are ready to multiply 5 digits, solve question 1a and 1c. For students who need additional practice with fewer digits, complete questions 1b and 1d.

1.4.3 Guided Instruction and 1.4.4 Guided Practice - Consider the area model as an entry point for visual learners and using arrows or color-coding the partial products within the area model to make specific connections to how they appear in the standard algorithm.

1.4.5 Try It – Consider having students create a math journal entry or interactive notebook page to solve each of three problems using one of the three methods. Consider labeling each method, creating an anchor poster on chart paper, and/or using this work as a visual reference for correctly used algorithms and models for students in later lessons.</td></tr>
<tr><td colspan="5">Lesson Notes:</td></tr>
</table>

1.4.1 Warm-Up: Decimal Products

Component Overview: *Students use numbers 1 through 9 to write two factors to the tenths place that have a product as close to 50 as possible. Consider writing various student responses on a class whiteboard or shared space for an opportunity to review operations with decimals, place value, and rounding.*

1.4.1 Warm-Up: Decimal Products

1. Place a digit in each blue box so that the product is as close to 50 as possible. Use only digits 1 to 9, at most one time each.

$$\boxed{}.\boxed{} \times \boxed{}.\boxed{} =$$

Answers will vary.
Possible answers include: 6.1×8.2; 9.8×5.1

If students are unsure where to start, consider having students try a few different numbers that they created to notice how the values change. Use informal observations during this Warm-Up to determine entry points for students as they progress through the lesson.

1.4.2 Refresher: Reasoning With Powers of Ten

Component Overview: *Students will be working in partners to reason with powers of 10 and fractional thinking as it relates to determining decimal place value when multiplying decimals. Consider using additional scaffolding using place value charts or other strategies to clearly make the connection between place value and powers of 10. The modeled thinking in this Refresher may also be helpful to revisit later in the lesson for any students who have difficulty placing decimals in products.*

1.4.2 Refresher: Reasoning with Powers of Ten

Several expressions are shown.

$250 \cdot 0.01$	$250 \div 100$	$250 \cdot 0.001$	$250 \cdot \frac{1}{100}$	$250 \cdot 0.1$
$250 \div 1000$	$250 \cdot \frac{1}{10}$	$250 \cdot \frac{1}{1000}$	$250 \div 10$	

1. Work with your partner to complete the table by writing the expressions that match each value indicated.

Value	Equivalent Expressions		
25	$250 \cdot 0.1$	$250 \cdot \frac{1}{10}$	$250 \div 10$
2.5	$250 \cdot 0.01$	$250 \cdot \frac{1}{100}$	$250 \div 100$
0.25	$250 \cdot 0.001$	$250 \cdot \frac{1}{1000}$	$250 \div 1000$

2. When working together through the problem, Cheryl and Miranda had a disagreement.

Cheryl said that $\frac{1}{100}$ is equivalent to 0.001 because both have two zeros.	Miranda said that $\frac{1}{100}$ is equivalent to 0.01 because of place value.

a. Which student do you agree with?
Answers will vary.

b. Explain why you agree.
Answers will vary.

As an entry point for visual learners, consider writing the whole number factor and final product on a place value chart, and asking students to notice and wonder the difference between them.

For added complexity, consider asking students to provide a diagram, model, or other rationale as evidence for why they agree in question 2b. Options could include modeling fraction and decimal equivalence on a place value chart or using place value chips.

1.4.2 Refresher: Reasoning With Powers of Ten (continued)

Recall that fractions like $\frac{1}{10}$, $\frac{1}{100}$, and $\frac{1}{1000}$ can be used to find the location of the decimal point in terms of place value when multiplying decimals.

For example, 24×0.02 can be thought of as $24 \times \frac{2}{100}$ or $24 \times 2 \times \frac{1}{100}$. This is equal to $\frac{48}{100}$ which is read, "48 hundredths" and written as 0.48 in decimal form with the 8 in the hundredths place.

$$24 \times 0.02 = 0.48$$

3. With your partner, complete the equations and statements below to show how similar thinking can be used to find the products of decimals.

   a.

$$(0.7) \cdot (0.09) = (7 \times \frac{1}{10}) \cdot (9 \times \frac{1}{100}) = (7 \cdot 9) \cdot (\frac{1}{10} \cdot \frac{1}{100}) = \boxed{63} \cdot \frac{1}{\boxed{1000}}$$

   Written as a decimal, $(0.7) \cdot (0.09) =$ 0.063.

   b.

$$(1.5) \cdot (0.43) = (15 \times \frac{1}{10}) \cdot (43 \times \frac{1}{100}) = \boxed{645} \cdot \frac{1}{\boxed{1000}}$$

   Written as a decimal, $(1.5) \cdot (0.43) =$ 0.645.

Have students use correct vocabulary when naming decimals and fractions by place value to help strengthen connections for converting fractions and decimals. For example, for question 3a, ask students to read aloud 0.7 as "seven tenths," and explicitly make the connection to writing $\frac{7}{10}$ as a fraction.

Predicting estimations could be a helpful strategy for students' self-reflection and verifying solutions. Consider the following probing questions to position students to exercise these skills:

- *How does the size of the product compare to the size of the factor when the factor is multiplied by 0.1? 0.01? 0.001?*
- *Can you predict the outcome of multiplying 750 by 0.1 or 0.01 without calculating? If so, how?*

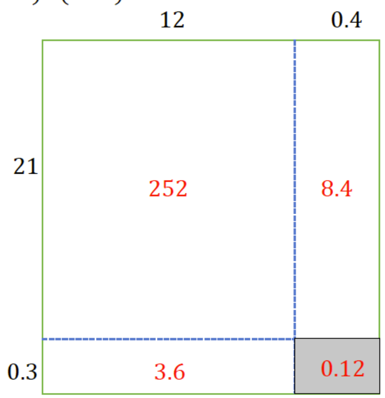
1.4.3 Guided Instruction: Multiplying Decimals to the Thousandths

Component Overview: *This section guides students through conceptual and procedural understanding to find the product of decimals to the thousandths place using three approaches: an area model, partial products, and the standard algorithm. The benchmark specifically identifies procedural fluency in the standard algorithm, but the lesson is designed to build on this appropriately based on prior learning.*

1.4.3 Guided Instruction: Multiplying Decimals to the Thousandths

Different strategies can be used to find the product of decimals to the thousandths.

1. Byanca prefers using an area model to find the product of decimals. Below is an area model that Byanca drew to represent $(12.4) \cdot (21.3)$.



a. Find the region that represents $(0.4) \cdot (0.3)$. Shade that region in the diagram and label it with its area of 0.12.
Answer shown in diagram.

b. Label the other regions of the model with their respective areas.
Answer shown in diagram.

c. Find the value of $(12.4) \cdot (21.3)$ using the labeled area model. Show your work.
$252 + 8.4 + 3.6 + 0.12 = 264.12$

Consider asking students the following:

- *Which parts of the factors would be multiplied to get 252?*
- *Which parts of the factors would be multiplied to get 8.4?*

1.4.3 Guided Instruction: Multiplying Decimals to the Thousandths (continued)

2. Sergio prefers to use partial products to find the value of $(12.4) \cdot (21.3)$, while Alicia prefers using the standard algorithm. Their work is shown.

Sergio's Work Partial Products	Alicia's Work Standard Algorithm
$$\begin{array}{r} 12.4 \\ \times\ 21.3 \\ \hline 0.12 \\ 0.6 \\ 3 \\ 0.4 \\ 2 \\ 10 \\ 8 \\ 40 \\ +\ 200 \\ \hline 264.12 \end{array}$$	$$\begin{array}{r} 12.4 \\ \times\ 21.3 \\ \hline 3.72 \\ 12.40 \\ +\ 248.00 \\ \hline 264.12 \end{array}$$

Discuss each method with your partner while answering the following questions.

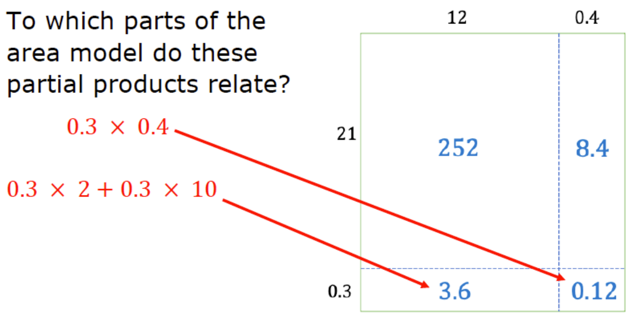
a. In Sergio's work, which parts of the factors were multiplied to get each partial product indicated in the table below?

Partial Product	Parts of Factors Multiplied
0.12	0.3 × 0.4
0.6	0.3 × 2
3	0.3 × 10

Notice and Wonder
Consider implementing a "Notice and Wonder" routine before students discuss and answer question 2.

- *Looking at these two approaches, what do you notice is the same and different about the two approaches?*
- *What parts of the students' work is similar?*
- *Describe which method you think is more efficient.*

b. To which parts of the area model do these partial products relate?



c. Where do the values of 3.72, 12.4, and 248 come from in Alicia's work that shows the standard algorithm?

- 3.72 is the product that results from multiplying by each part of 12.4.
  - ○ 0.3
  - ○ 1
  - ○ 20
- 12.4 is the product that results from multiplying by each part of 12.4.
  - ○ 0.3
  - ○ 1
  - ○ 20
- 248 is the product that results from multiplying by each part of 12.4.
  - ○ 0.3
  - ○ 1
  - ○ 20

d. Draw lines to indicate connections between the partial products found in each of Sergio and Alicia's methods.
Answer shown in table.

e. In each method, why are the numbers between the horizontal lines aligned vertically?
They are aligned vertically so the place values are aligned.

Using the nearest whole numbers of the factors, what is a good estimate for the product of these factors? How can I use this estimation to determine where to place the decimal point in the product of 12.4 and 21.3?

Notice that questions 2c and 2d position students to see explicit connections between the partial products method and the standard algorithm to build students understanding. Guide students to visual entry points using question 2d. As an entry point for auditory learners, verbalize the connection while drawing the arrows using phrases like "similar to" or "is the same as" to make connections.

Consider the entry point for question 2d by helping draw the lines on question 2a, then ask students to draw the lines individually in question 2b. This support may be helpful for visual learners to first see the connection before identifying it on their own.

1.4.4 Guided Practice: Multiplying Decimals to the Thousandths

Component Overview: *Students apply what they learned about multiplying decimals with the area model, partial products, and the standard algorithm to try each method with a partner. This practice guides students to identify errors, make corrections, and discuss students' opinions on the strengths/weaknesses of each method.*

1.4.4 Guided Practice: Multiplying Decimals to the Thousandths

1. Find the product $(32.5)(6.02)$ using the area model below. 195.65

	32	0.5
6	192	3
0.02	0.64	0.01

2. With your partner, determine who will be Partner A and Partner B.
   a. Find the product of 3.621×9.7 using the method indicated. Show your work in the space provided.

Partner A Partial Products	Partner B Standard Algorithm
3.621 $\times$ 9.7 0.0007 0.014 0.42 2.1 0.009 0.18 15.4 + 27 35.1237	3.621 $\times$ 9.7 25347 + 325890 35.1237

   b. Compare your work with your partner to verify that you both found the same product. Discuss any errors and make corrections as needed. Once you have come to an agreement, write your partner's work in your workbook in the space provided.
   c. Discuss with your partner the strengths and weaknesses of using each method.
   d. Which method do you prefer for multiplying decimals to the thousandths? Answers will vary.

Synthesis is critical at this stage. Consider how you are positioning students to navigate across methods and achieve fluency by selecting methods that work best for them.

- *Intentionally include time for comparing and making sense in question 2*
- *Purposefully pair students of different learning styles together*

If students are having difficulty with placement of decimal digits, avoid teaching shortcuts like counting the number of decimal places to determine the placement of the decimal in the product. Consider instead modeling fractional thinking or estimation to help students make conceptual sense of this.

1.4.5 Try It: Multiplying Decimals to the Thousandths

Component Overview: *Students apply a preferred method to multiply decimals up to the thousandths place. Use this section as an opportunity to focus on procedural fluency. Consider having students that used different methods use their preferred method to solve one of the problems.*

1.4.5 Try It: Multiplying Decimals to the Thousandths

1. Find each product using your preferred method. Show your work.

   a. $25.36 \cdot 14.8$
   375.328

   b. 19.008×3.2
   60.8256

   c. $(36.3)(0.74)$
   26.862

Consider a math journal entry, foldable, or type of interactive notebook page for this practice. Students could solve each of the questions 1a-c using one of the three methods from this lesson, then check answers with partners or the teacher for accuracy. This could provide a record of correctly used methods for students to reference in later lessons.

For additional support both in 1.4.5 Try It and 1.4.6 Your Turn!, consider offering Math Nation's On-Ramp video on Multiplying Decimals to students.

1.4.6 Your Turn!

Component Overview: *Students use any method to accurately multiply decimals to the thousandths place. Consider adapting by specifying which questions students attempt based on student needs and procedural fluency using various multiplication approaches. Questions 1a and 1c include factors with 5 digits, while questions 1b and 1d have factors with 2-4 digits.*

1.4.6 Your Turn!

1. Find each product.

   a. 105.46×3.8
   400.748

   b. $8.7 \cdot 0.475$
   4.1325

   c. $(7.7)(102.65)$
   790.405

   d. 1.561×12.8
   19.9808

Consider using this section to encourage students to use more than one method to build fluency and conceptual understanding of decimal multiplication across approaches.

How can you use estimation to determine if your product is reasonable?

1.4.7 Wrap-Up: Ticket Out the Door

Component Overview: *Consider using all or part of this fluency check as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' ability to fluently multiply decimals to the thousandths place. It does not require the use of a particular method. Students should not use a calculator to complete the Wrap-Up.*

1.4.7 Wrap-Up: Ticket Out the Door

Encourage students to use the method of solving that makes the most sense to them. It may be helpful to provide a blank area model for students to fill in.

1. Find each product.

   a. $(6.36)(12.27)$
   78.0372

   b. $8.12 \cdot 29.4$
   238.728